

## **Practical Assignment 1 - Exploring Protein Sequence and Structure**

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BIOS331: Microbial Genomics



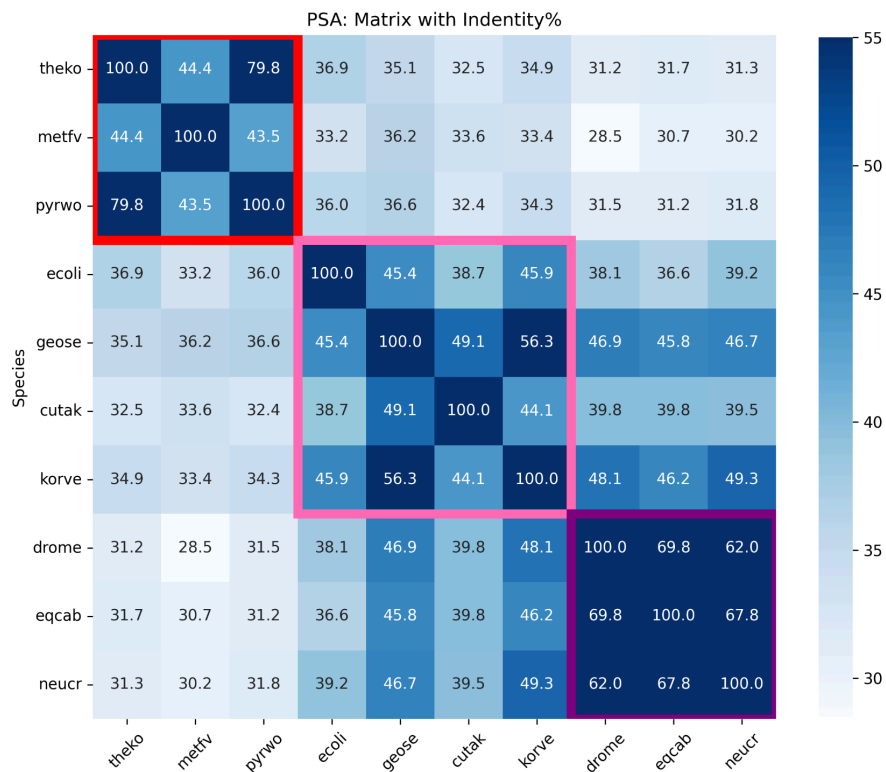
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## Introduction

For this assignment, I've considered the sequence of Phosphoglycerate kinase (PGK), an important enzyme implicated in Glycolysis. It is well documented that PGK is conserved across all three taxonomic domains, i.e., Archaea, Bacteria, and Eukarya. This report studies sequence similarity and evolutionary distance across PGK homologues. The assignment considers sequence homologues from 10 species, across the three taxonomic domains. Throughout the scripts, for succinctness, I use shortened labels for each species, which are mentioned in the brackets. For Archaea, I've considered *Thermococcus kodakarensis* (theko), *Methanothermus fervidus* (metfv) and *Pyrococcus woesei* (pyrwo). *Escherichia coli* (ecoli), *Geobacillus stearothermophilus* (geose), *Cutibacterium acnes* (cutak) and *Koribacter versatilis* (korve) are considered for Bacteria. The representatives for Eukarya are *Drosophila melanogaster* (drome), *Equus caballus* (eqcab) and *Neurospora crassa* (neucr). The protein sequences were collected through UniProt's programmatic API access (here: <https://www.uniprot.org/help/api>). Throughout this write-up I will refer to the code written in an Ipython Notebook, as well as other pertinent files that will be shared via Google Drive.

## Pairwise Sequence Alignment (PSA)

On using EMBOSS-NEEDLE (Rice et al. 2020, Madeira et al, 2024) for species-wise PSA, and inputting the pertinent sequences in each run on the web interface, the following identity% matrix is received:



**Fig 1: PSA Identity% Matrix.** Box outlines represent the domain-domain PSA identity%. Red box is for Archaea-Archaea, Pink is for Bacteria-Bacteria, and Purple is for Eukarya-Eukarya alignments. Notice how intra-domain identity % is higher compared to cross-domain PSAs

Below is an example of a EMOSS-NEEDLE output:

```
#=====
# Program: needle
# Rundate: Wed 30 Jul 2025 18:50:06
# Commandline: needle
# -auto
# -stdout
# -asequence emboss_needle-I20250730-184957-0160-94143339-p1m.asequence
# -bsequence emboss_needle-I20250730-184957-0160-94143339-p1m.bsequence
# -datafile EBL0SUM62
# -gapopen 10.0
# -gapextend 0.5
# -endopen 10.0
# -endextend 0.5
# -aformat3 pair
# -sprtein1
# -sprtein2
# Align_format: pair
# Report_file: stdout
#####

#=====
#
# Aligned_sequences: 2
# 1: EMBOSS_001
# 2: EMBOSS_001
# Matrix: EBL0SUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
# Length: 425
# Identity:      141/425 (33.2%)
# Similarity:   213/425 (50.1%)
# Gaps:         53/425 (12.5%)
# Score: 529.0
#
#
#=====

EMBOSS_001          1 MFKFYTMDDFDYSGSRVLVRVDINSPVDPHTGRILDDTRMLRHSKTLKEL           50
    .....|.||.:|.|:||.:|.|:||..|:::..|.|.:|.....|:::
EMBOSS_001          1 -MSVIKMTDLDLAGKRVFIRADLNVVP---KDQKVTSNDARIRASLPETIELA       47
EMBOSS_001          51 VDENAKVAIIAHQSRRPGRKDFTTMEEHS-----KVLSTNIDMPPTYVEDI        95
    :|::|.|.||.:|.|.:|.....| | | | ..|:::..|.|.:|.....|
EMBOSS_001          48 LKGAGAKVMVTSHLRPTGEYN---EESFLPLVVNIKKDLNSNPVRLVKDY          95
EMBOSS_001          96 FGCAARESIRNMENGDIILLLENRVFYSEEVLRDPRPKVQAETHLVRLKLSV       145
    .... .:::|:::| |||||...| |:|::: .| |.:|:::
EMBOSS_001          96 LDGV-----DVAEGELVPLENVRFNKEG---KKDET-----LSKKYAAL       132
EMBOSS_001          146 VDYYINDAFAAHRSQPSSLVG-F-PLKLPSAAGRLMERVKTYKIKNVE          194
    .|:::| |||. | | | |.:|.|.:|.....| | | | |:::~| |.:|.....|
EMBOSS_001          133 CDVFMDAFGTGAHQASTHGIGKFADVACAGPLLAAELDALKALKEPA            182
EMBOSS_001          195 KPCVYLGGVKIDDSIMIMKNILKNGSADYILTSLGVANVFLEASGDIK           244
    :|.||.:|.|.:|.|.:|.....| | | | |:::~| |.:|.....| | |.:|
EMBOSS_001          183 RPMVAIVGGSKYSTKLTVLDSLK---IAQLIVGGGIANTFIAAQGHV-             229
EMBOSS_001          245 EKNRKILYRKMYKKFIKMAKKLKDKYGEKITLPDVAVINKNGKRI-----       289
    .| | | | |:::~| |.:|.....| | | | |:::~| |.:|.....|
EMBOSS_001          230 ---GKSLEYAD---LVDEAKRLTTC--NIPVPSDVRVATEFSETAPATL          271
EMBOSS_001          290 ----DVPIDDIPNFPIYDIGMETIKIYAEKIREAKTI FANGPAGVFEEQQ          335
    | |.:| | | | | | | | | | | | | | | | | | | | | | | | | | | |
EMBOSS_001          272 KSVNDVKADE----QILDIGDASAQELAETLNKAKITLWNGPVGVFEFPN          317
EMBOSS_001          336 FSIGTEDLLNAIASSNAFSVIAGGHLLLLAAAAEKMGISNKINHISSGGGACI     385
    | | | | |.:| | | | | | | | | | | | | | | | | | | | | | | |
EMBOSS_001          318 FRKGTIVANAIAOSEAFSIAGGGDTLAIDLFGIADKISYISTGGGAFL          367
EMBOSS_001          386 AFLSGEELPAIKVLEEARKRSKYI              410
    .|.||.:|.|.:|.|.:|.....|
EMBOSS_001          368 EFVEGKVLPAVAMLEERAKK-----              387

#-----
#
```

**Fig 2: Example output from EMBOSS Needle for the sequence alignment of *Methanothermus fervidus* and *Escherichia coli*. Notice how the identity percent corresponds to the matrix in Fig 1.**

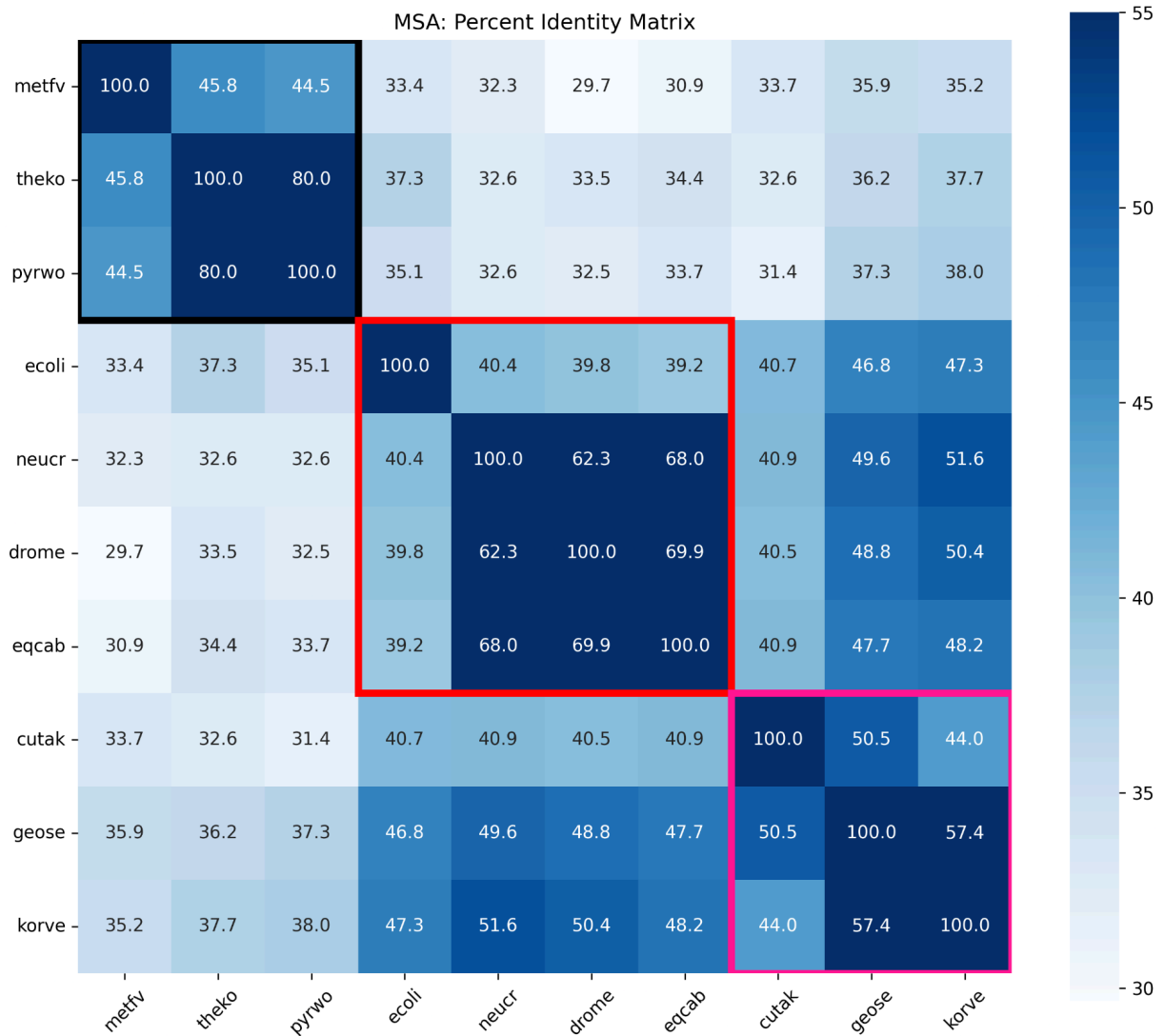
## Multiple Sequence Alignment (PSA)

For MSA, I use Clustal Omega (Rice et al. 2020, Madeira et al, 2024), and AlignmentViewer (Reguant et al., 2020) to visualize the alignment, and subsequent conservation analysis. The input to Clustal Omega was a fasta-like text input generated programmatically in the Python Notebook referred. The output of Clustal Omega was a Fasta file, displayed here:

```
>metfv
---MFKFTYMDDFDYSGSRVLVRVDINSPVDPHTRILDDTRMLHSLTKLVDENAKV-
ATLAHOSRPGKRDFT---TMEHSHKLSNILD-----MPVTYVEDIFGCAARESRNME
NGDIILLENVRFYSEEVLRDP-----KVOAETHLRKLSVVDYINDAFAAHRS
QPSLVGFPLKLP5--AGRLMEREVKTLKIKNVEKPCVYILGGVKIDDSIMIMKILKN
GSADYILTSGLVANVFLEAS--GIDIKENRKILYRKNYKFIKMAKKLKDYGEKILTPV
DVAINKNGKRIDV---PID---DIPNFPYIDIGMETIKIYAEKIREAKTIFANGPAG
VFEEQQFSIGTEDLLNAIAS---SNAFSVIAGGHLAAAEKMGISN-KINHISGGGGACI
AFLSGEELPAIKVLEAKRSDKIYI-
>theko
---MFRITDFTYEGKTVFLRADLNSPVE---KGRITSDFRAVLPTIQHLNDGAKL-
VIATHOSRPYGDYI---TTEHAQILSRLLG-----QEVEYEDIFGKYARERISLK
PGEAIILENLRFSAEEVNRSI---EECEKTFVRKLLAPLDYVNDAFATAHRS
QPSLVGFARLKP5--IMGKLMETEDALSKAYSEERPRVYLGAKVDDSLRAENVLRK
GKADLILTGLVQGITFLAK--GFDLGDAMIEFLHRKGLLELDVDAEKILDEFYYPVTRPV
DFAIDYGERLEV---DLLGGEKRLFDQYPLIDIGSRTEKYREILIGAKTIIVANGPMG
VFEREFAVGTGVFRAIGE---SPAFSVGGGHSIASIYRNYIT---GISHISTGGGAML
SFFAGEELPVKALKISYERFKDRVKE
>pyrwo
---MFLRDFEYNNRTVFLRVLDNSPMS---NGKITSDFRAVLPTIKYLTESGAKV-
VVGTHQGPYSEYS---TTEHARILSELLN---MHVEYEDIFGKYARERIKAMK
PGEVIVLENLRFSAEEVKNATI---EECEKTFVRKLSQVLDVNDAFAAHRS
QPSLVGFARLKP5--IMGFLMEKVDALTKAYSEKPRVYLGAKVDDSLKVAENVLRK
EKADLILTGLVQGLFTLAK--GFDLGRNIEKFLKKGILKYVDWAEEKILDEFYYPVTRPV
DFAIDYGERVEI---DLLSDEKRLFDEYPLIDIGSRTEKYREILIKARIIVANGPMG
VFEREFAVGTIGVKAIGE---SPAFSVGGGHSIASIYKYNIT---GISHISTGGGAML
TFFAGEKLPVLEALKISYEKSNLLS-
>ecoli
---MSVIMKTDLDLAGKRVFIRADLNPVVK---DGKVTSDARISLPTIELALKQGAKV-
MVTSHLRPTEGEYNEEFSLLPVWYLYKDKLSNPVRLVKDYL-----DGVDVA
EGELVVLNVRFNKGEKKD---D---ETLSKKYAAALCDVFMDFAGTAHRA
QASTHGIGKFDVACAGPLLAELDALGKALKEPARPMVAIVGGSKVSTKLTVLDSLKI
---ADQLVGGGIANTFIAAQ--GHDVGKSLYEADLVDEAKRLTT-----CNIPVPS
DVRVATEFSETAPA---TLKSVNDVKADEQILDGDASQELAEILKNKATILWNGPVG
VFEPNFRKGTEIVANAID---SEAFSAGGGDTLAAIDLFGIAD--KISYISTGGGAFI
EFVEGKVLPAVAMEERAKK-----
>neucr
MSLSNKLSDIEDVLKGRVLRVDFNVPLDA-EKKVTNPQRIAGAIPTIKYALDHGAKAV
VLMSHLGRPDGKP-NPKYSLKPVVPELEKLLG---KQVTFAPDCVGPVEEIVNKAD
NGEVIILLENLRFTHIEEGKGTDAEGNKVKADKAKVEFRNRLTKLDYVINDAFGTAHRA
HSSMVGTD---LPYKAAGFLMKELDYFAKVLSPORPFLSTILGGAKVSDKIQLINLLDK
---VNTLIVCGGMATFKKLTQNMPTIGSLFDEAGAKIVPDLVYKAEK---NNVKLVLVP
DFTIADKFDKANT---GYATDKDGIPOGMGLDCGEESVKLFTQAINESOTILWNGPAG
VFEDKFAKGTATLDACVKAEEGRTVIIGGDTATVAAKYGVED--KLSHVSTGGGASL
ELLEKALPGVVALSERQ-----
>drome
---MAFNKLSIENLDLAGKRVLMRVDFNVPKE---GKITSNQRIVAALDSIKLAKSKAKSV
VLMSHLGRPDG--NKNIKYTLAPVAAELKTLG---QDVFLSKDCVGEVEEACKDPA
PGSVILLENVRFYVEEKGKLDASGGKVKADPAKVEFRASLAKLDYVINDAFGTAHRA
HSSMVGTD---FEQRAAGLLNKLKYFQALDPPNPFILAIKGGKAVADKIQLINLLDK
---VNEMIIGGGMATFLKVLNNMIEIGTSLFDEEGAKIVKNSKAKEK---NGVKITLVP
DFVCGDKFAENAAV---SEATVEAGIPDGHMGLDVGPRTRELFAPIARAKLIVWNGPVG
VFEPNFMANGTSLMDGVVAATKNGTSLIIGGDTATCCAKMTEA-LVSHVSTGGGASL
ELLEKTLPGVAALTS-----
>egcab
MSLSNKLTLDKLNVKGRVLMRVDFNVPKNI---NQITNNQRIKAAVPSIKFLDNGAKSV
VLMSHLGRPDGVPMPDKYSLQPVAVELKSLG---KDVFLSKDCVGPVEEACKADPA
AGSVILLENLRFHVEEKGKLDASGKVKAEPAKTIETFRASLKLGDYVINDAFGTAHRA
HSSMVGVD---LPQKAGGFLMKKELNYFAKALSPERPFILAIKGGKAVADKIQLINLLDK
---VNEMIIGGGMATFLKVLNNMIEIGTSLFDEEGAKIVKNSKAKEK---NGVKITLVP
DFVTADKFDENAKT---GQATVASGIPAGMGLDCGTESSKYYAEAVARAKQIVWNGPVG
VFWEAFARGTKALMDEVVKATSRGCTIIGGDTATCCAKMTEA-LVSHVSTGGGASL
ELLEKVLPGVDALSNV-----
>cutak
---MKSVEDLDLGRKRVIRCDNFVPLD---GDKITDDGRKIAALPTLTQLHEAGART-
TVLAHLGRPKGAV-DPKYSLAPYAKRLGELMNLVDKL---AGDIVGPSATKTSIESLA
DGETALVENVRFEKGETSK---DEAERAEALQKYASGFEPPYNSGFGVWHRK
QASVYDVAKLLP5--AGSLVAKVEVELGLTADPKRPFVYLGAKVADKLAVIDLKKA
---ADTLVIGGGMATFLKAK--GLEVGDSLLDESDIDACTKYLEQAKA---AGKQILLPV
DVRVVSIDIDFEARTVGQIDVVPADIEPSDKMAVIDGPETEKAYANAIGKASVFWNGPMG
VFETKLELGTAKAQAALTE--V-DGLSVVGGDSASAVRTLGFADDQFGHISTGGGASL
ELMEGKELPGI AVLKENN-----
>geose
---MNKKTIRDVDRGRKRVIRCDNFVPMPE---QGAITDDTRIRAALPTIRYLIEHGAKV-
ILASHLGRPKGV--VEELRLDAVAKRLGELLERPVAK-----TNEAVGDEVKAAVDRNL
EGDVLLENVRFYPGEEKN---D---PELAKAFELADLYNDAFGAHRA
HASTEGIAHYLPA--VAGFLMEKEVLGKALSNPDRPPTAIGGAKVKDKIGVIDNLLK
---VDNLIIGGGGLATYFKAL--GHDVGKSLLEEDKIELAKSFMEKAKE---KGVRFYMPV
DVRVADRFDANDANT---K--VVPIDATPADWSALDIPKTRILYRDYIREKLVWNGPMG
VFENDAFAGTKATAEALAE---ALDYSYVIGGDSAAAEKFGGLAD--KMDHISTGGGASL
EFMEGKQLPGVVALEDK-----
>korve
---MSKLSIKDLQLSNKRVFMRVDFNVPLDE--NGRVTDTRITRELTPIEYALRHGAKL-
ILCSHLGRPKGP--NPKMSLKPAERLRVMDHAI5PGONVGFSPDCIGMAOEMAKOLE
KGOALLLENVRFHAEFEKN---D---PAFAKELASLCELVNDAFGSAHRA
HASTEGITHYVEKSAAGLLMQKELDYLKATSNPAKPFVAILGGAKVSDKIGVIONLMAK
---VDAILIIGGGMATFLKAQ--GQEGKSLFEADKLDAKILADAHK---RGLKFLLPV
DHVTADKDFMDHATP---HQIGEGQIPAEQMALDIPKTVLFSSEIAKARTIVWNGPMG
VFEDNFAKGTARAIKAVAG--NSGATSVGGGDSVAAVHDGVAO--KITHISTGGGASL
EFLGKLPVGEALTNK-----
```

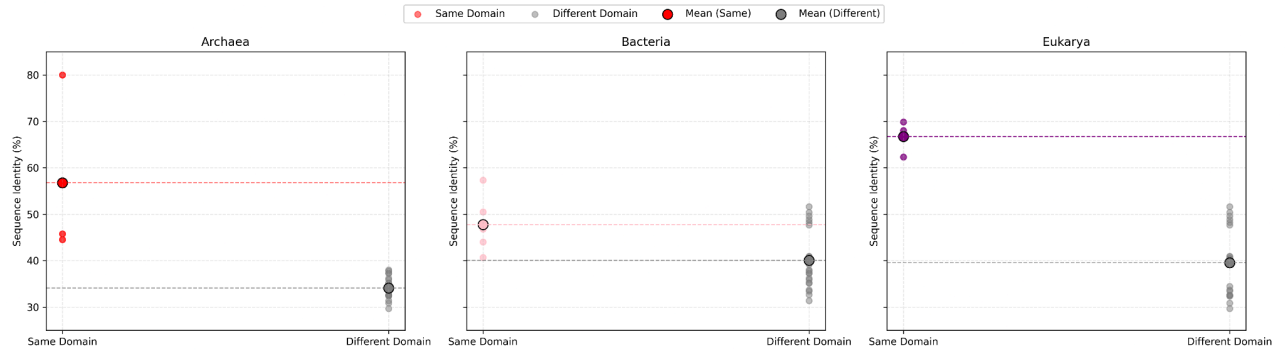
Fig 3: Clustal Fasta Output on Multiple Sequence Alignment

We can observe a similar identity matrix when we consider the MSA Percent Identity matrix obtained from Clustal.



**Fig 4: MSA Identity% Matrix. Box outlines follow the same schema as in Fig 1.**

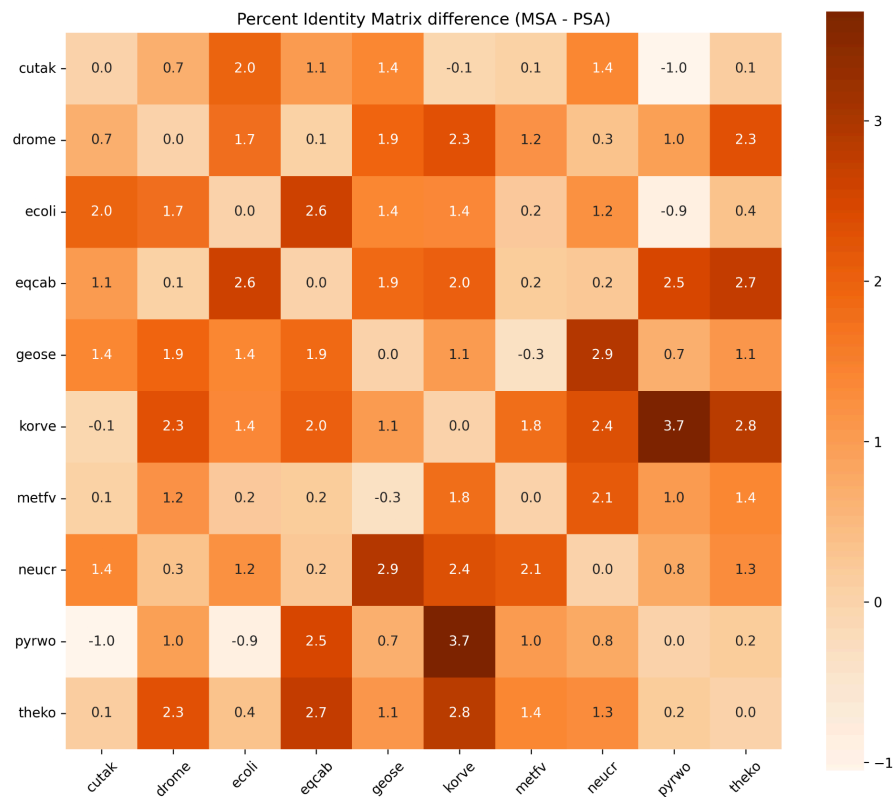
The identity % continues to be higher in these boxes compared to cross-domain MSA identity %, consistent with the observations in Fig 1. To further validate this, we can compare PGK sequence identity between species from the same domain and those from different domains, as in the following figure:



**Fig 5: Dot plot for sequence identity within and across domains. Colored dots show intra-domain comparisons; gray dots show cross-domain comparisons. Dashed lines indicate domain means.**

We can see that each domain shows markedly higher intra-domain identity than compared to cross-domain identity. This reinforces the idea of evolutionary conservation within taxonomic groups.

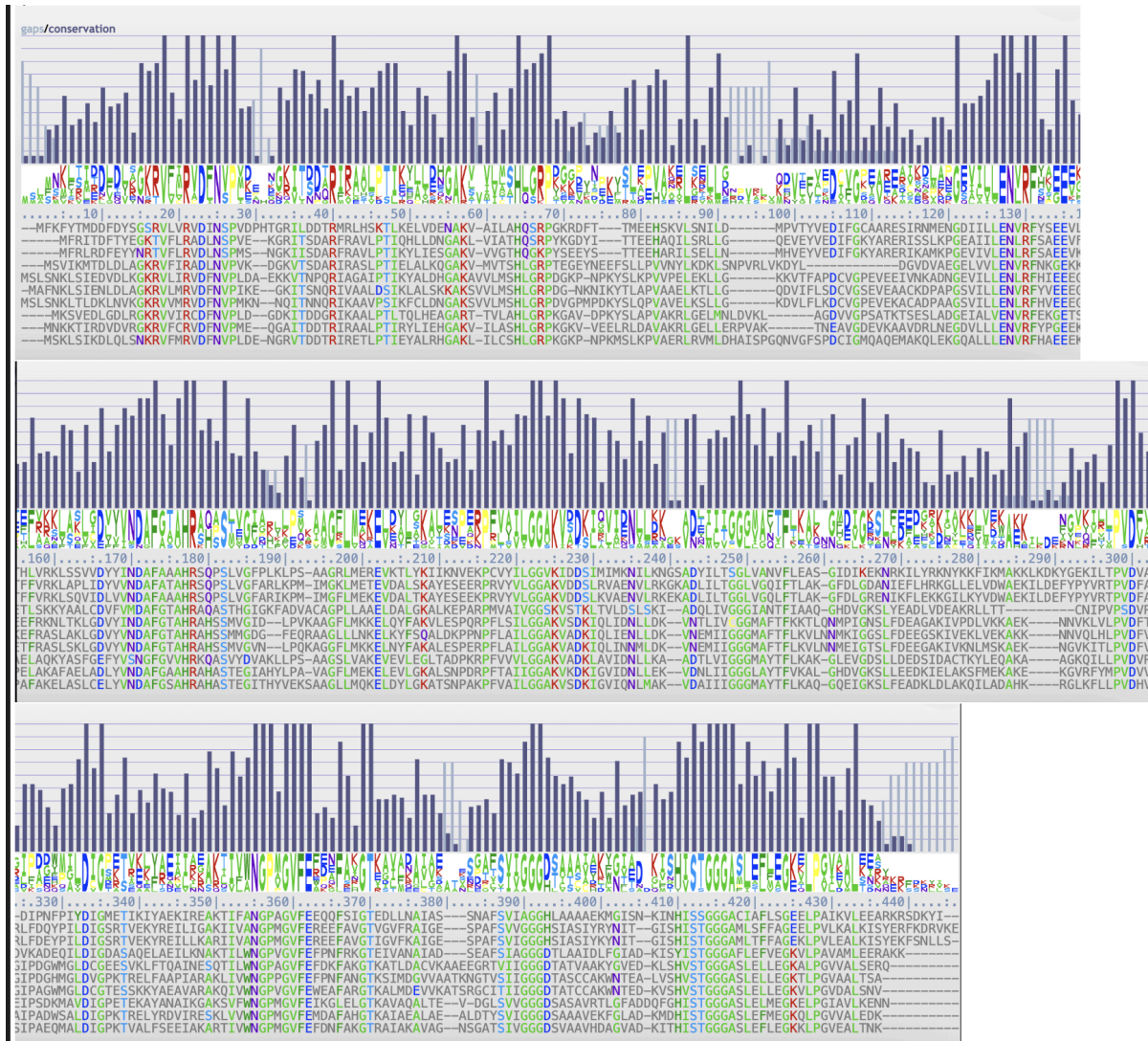
Also, it is noticed that there isn't a significant difference between the Matrices in Fig 1 and Fig 4, as shown by this difference heatmap here:



**Fig 6: MSA - PSA difference matrix shows no great differences in MSA and PSA results.**

## Conservation Analysis

For conservation analysis, I use AlignmentViewer to highlight the residues conserved in  $\geq 70\%$  of the sequences (i.e., present in at least 7 out of 10 aligned sequences). Thresholds lower than 70% had a lot of noise. This is the order of the sequences according to which AlignmentViewer visualises the alignments: metfv, theko, pyrwo, ecoli, neucr, drom, eqcab, cutak, geose and korve. The figure below stitches different parts of the visualisation to give a comprehensive view of all the conserved sequences.



**Fig 7: AlignmentViewer output.** This visualisation shows only those segments of the sequence that contained conserved residues.

As a mentioned aim of this analysis was to understand the relationship between sequence similarity and evolutionary distance, I used EMBOSS distmat (Rice et al. 2020, Madeira et al. 2024) to construct distance matrices, choosing the Jukes Cantor correction, inputting the MSA Fasta file. This was the output:

**OUTPUT FILE** [outfile](#)**Distance Matrix**

Uncorrected for Multiple Substitutions  
Gap weighting is 0.000000

| 1    | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |          |
|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|
| 0.00 | 87.80 | 20.00 | 92.76 | 92.64 | 82.09 | 88.83 | 92.94 | 94.40 | 94.16 | theko 1  |
|      | 0.00  | 87.80 | 91.99 | 92.64 | 92.79 | 82.63 | 94.63 | 93.66 | 92.93 | metfv 2  |
|      |       | 0.00  | 91.99 | 92.64 | 83.08 | 88.83 | 91.95 | 94.88 | 94.39 | pyrwo 3  |
|      |       |       | 0.00  | 85.01 | 94.32 | 89.41 | 90.18 | 92.51 | 92.76 | ecoli 4  |
|      |       |       |       | 0.00  | 95.18 | 89.34 | 94.16 | 91.37 | 92.64 | geose 5  |
|      |       |       |       |       | 0.00  | 92.54 | 93.03 | 93.03 | 92.79 | cutak 6  |
|      |       |       |       |       |       | 0.00  | 94.04 | 92.31 | 92.56 | korve 7  |
|      |       |       |       |       |       |       | 0.00  | 93.49 | 95.42 | drome 8  |
|      |       |       |       |       |       |       |       | 0.00  | 37.41 | eqcab 9  |
|      |       |       |       |       |       |       |       |       | 0.00  | neucr 10 |

**Fig 8: Output of Emboss-Distmat showing the evolutionary distance matrix.**

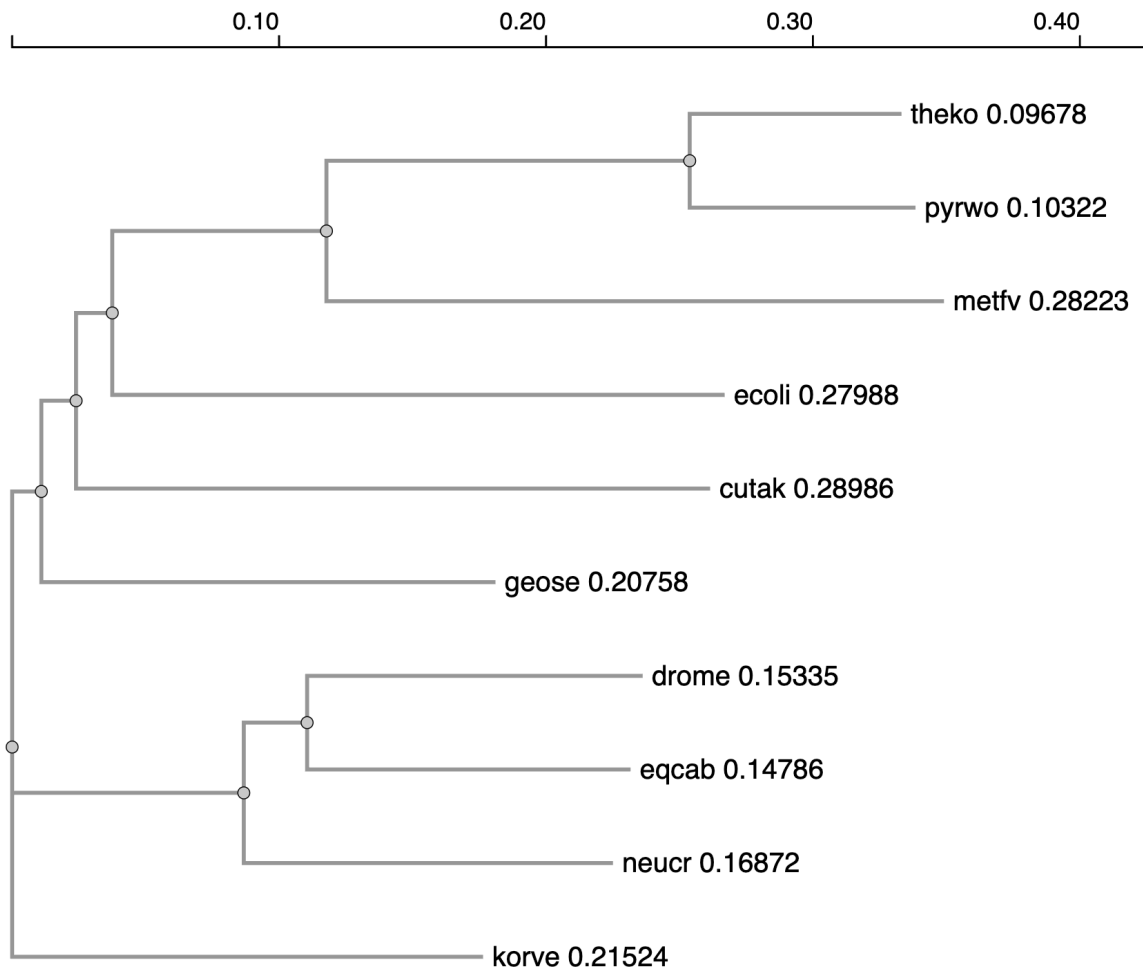
On calculating the Pearson correlations between each species' pairwise distances (from the distance matrix) and its corresponding % identity values from the MSA (excluding self-comparisons), we can construct this table:

|    | Domain   | Species | Correlation (r) |
|----|----------|---------|-----------------|
| 1  | Archaea  | theko   | -0.993          |
| 2  | Archaea  | metfv   | -0.998          |
| 3  | Archaea  | pyrwo   | -0.992          |
| 4  | Bacteria | ecoli   | -0.998          |
| 5  | Bacteria | geose   | -0.997          |
| 6  | Bacteria | cutak   | -0.996          |
| 7  | Bacteria | korve   | -0.997          |
| 8  | Eukarya  | drome   | -0.991          |
| 9  | Eukarya  | eqcab   | -0.992          |
| 10 | Eukarya  | neucr   | -0.993          |

**Fig 9: Correlation between pairwise evolutionary distance and MSA identity (%).**

The results show strong negative correlations ( $r \approx -0.99$ ) for all species, across all domains. This highlights that generally, as sequence identity increases, evolutionary distance decreases.

I also used EMBL-EBI's Simple Phylogeny tool (Madeira et al, 2024) to construct a phylogenetic tree from the MSA fasta file. Default parameters and heuristics were used to obtain the following tree:

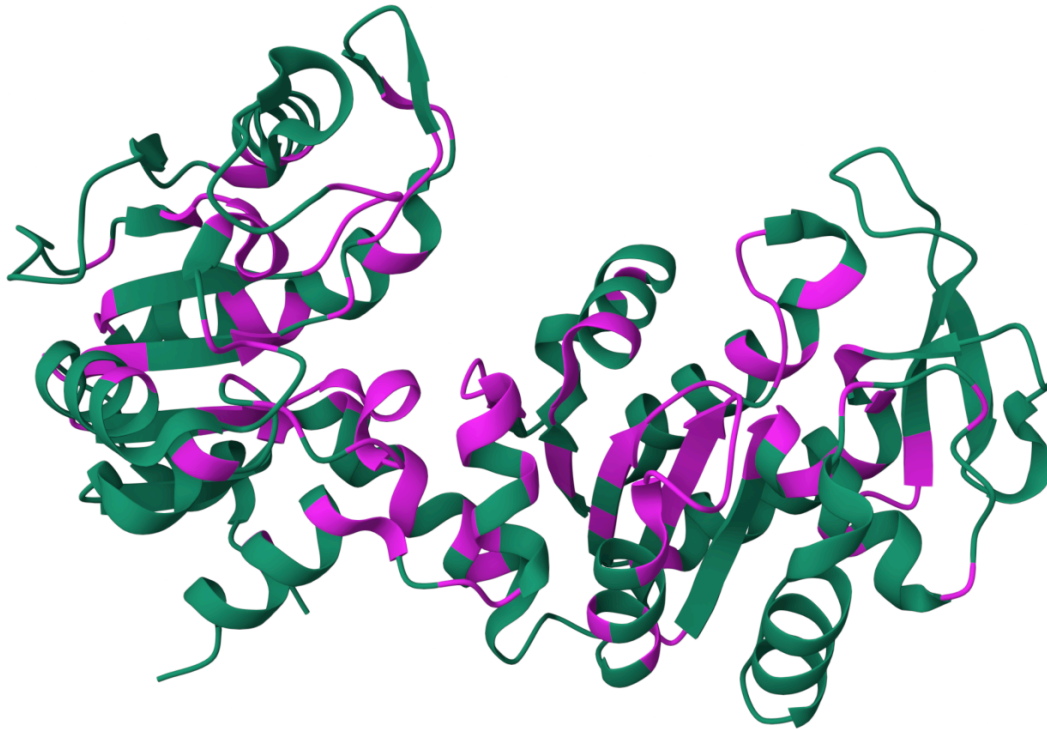


**Fig 10: Phylogram generated by Simple Phylogeny. The numbers on the image represent a measure of inferred phylogenetic distance.**

The phylogenetic tree clusters species from Archaea, Bacteria, and Eukarya based on PGK sequence. While most taxa group according to their domains, *Koribacter versatilis* (korve) appears as an outgroup. A possible reason is that despite being bacterial, *K. versatilis* belongs to a deeply branching phylum. Its placement likely reflects genuine phylogenetic distance from other bacterial representatives rather than misclassification. It is also worth noting that this topology is based on a single protein sequence, and thus observed divergence reflects protein-level variation and may not directly correspond to taxonomic divergence.

## Mapping Conserved Residues onto 3D protein structure

For this part of the assignment, I consider the *E. coli* PGK protein structure, as the crystal structure was readily available on RSCB/PDB (Marqusee, 2005). I annotated all sites in RSCB's Mol\* (Sehna et al. 2021) that were conserved in >70% of the sequences considered in pink, based on the AlignmentViewer output. This is visible in this figure:



**Fig 11: *E. coli* PGK crystal structure (X-ray diffraction) annotated with sites conserved in  $\geq 70\%$  of the sequences**

Out of the 387 amino acids in *E. coli* PGK, 117 residues were conserved in  $\geq 70\%$  of the sequences analyzed, corresponding to  $\sim 30\%$  overall conservation. Of these, 70 were located within  $\alpha$ -helices or  $\beta$ -sheets, yielding a ratio of 70:47 for conserved residues in structured elements versus loops. This implies a nearly 1.5-fold enrichment of conserved residues in secondary structural regions compared to unstructured regions.

This pattern of enrichment is consistent with expectations, as  $\alpha$ -helices and  $\beta$ -sheets generally constitute the structural core of proteins and play a critical role in maintaining the functionality of the enzyme. The catalytic mechanism of PGK involves a large conformational transition (Shirakihara & Evans, 1988), meaning that structural integrity within protein domains is therefore likely essential for this dynamic mechanism. These constraints, especially pronounced in enzymes like PGK that are essential to central

metabolism, subject such residues to strong selection pressure. Consequently, we can infer that conserved residues within  $\alpha$ -helices and  $\beta$ -sheets are hence associated with functional roles.

## References

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